Indian Institute of Engineering Science and Technology, Shibpur

B. Tech (CST), 3rd Semester, Mid-Semester Examination, September, 2024

Discrete Structures (CS2101)

Full Marks: 30 *Time: 2 Hours*

- *Answer only three questions*
- *No answer to extra question will be evaluated.*

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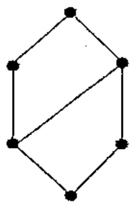
1. Choose the correct alternative. No explanation is needed for your answer.

(a) If $U = \{a,b,c,d,e,f,g,h\}$, $A = \{a,b,c,g\}$, $B = \{d,e,f,g\}$ then $\overline{A} - B$ is
(i) $\{g,h\}$ (ii) $\{h\}$ (iii) $\{d,e,f\}$ (iv) Φ

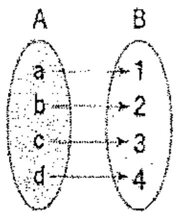
(b) $\neg(p \to q)$ is same as
(i) $p \vee \neg q$ (ii) $\neg p \vee q$ (iii) $p \wedge \neg q$ (iv) $p \vee \neg q$ where $p,q \in \{0,1\}$

(c) If $(D_n, /)$ is a poset, D_n: divisors of n then what is the value of n for the following Hasse diagram? Here, labels of the diagram are not provided intentionally.

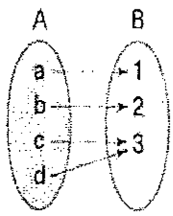


(i)16 (ii)12 (iii)20 (iv)10

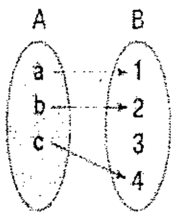
(d) Which of the following is injective function?



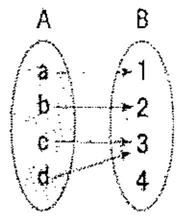
f_1



f_2



f_3



f_4

(i) Only f_3 (ii) Both f_1 and f_3 (iii) Only f_1 (iv) Only f_4

(e) In a relation $R = \{(p,1),(q,3),(r,2),(s,4)\}$, then $a \notin R^{-1}$ is
(i) (4, s) (ii) (3, q) (iii) (2, r) (iv) (3, p)

(f) If $f(x) = 2x+3$, $g(x) = x^2-1$ then $(fog)(x)$ is
(i) $2(x^2-1)+3$ (ii) $2(x^2-1)-3$ (iii) $(x^2-1)+3$ (iv) $2x^2-x+3$

(g) Which of the following statements about the function $f(x) = x^2-4x+3$ is true?
(i) The function has no real roots.
(ii) The function is increasing for all values of x.
(iii) The function can be factored into $(x-1)(x-3)$.
(iv) The function is a linear function.

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(h) Which of the following statements is a tautology?

(i) $(P \wedge Q) \vee \neg (P \vee Q)$

(ii) $P \vee \neg P$

(iii) $(P \rightarrow Q) \wedge (Q \rightarrow R)$

(iv) $P \wedge (Q \vee \neg P)$

(i) To prove the statement "If x is an odd integer, then x^2 is odd," which of the following is a correct direct proof?

(i) Assume x is odd, so $x = 2k+1$ for some integer k, Then $x^2 = (2k+1)^2 = 4k^2 + 4k + 2$ which is odd. So x^2 is odd.

(ii) Assume x is odd, so $x = 2k+1$ for some integer k, Then $x^2 = (2k+1)^2 = 2(2k+1)$ which is odd. So x^2 is odd.

(iii) Assume x is odd, so $x = 2k+1$ for some integer k, Then $x^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ which is form of $x = 2m+1$ where m is an integer. So x^2 is odd.

(iv) Assume x is odd, so $x = 2k+1$ for some integer k, Then $x^2 = (2k+1)^2 = 4k^2 + 4k + 1$ which is even. So x^2 is odd.

(j) Given a set S where $|S| = 12$, how many subsets of S have exactly 3 elements?

(i) 220 (ii) 792 (iii) 924 (iv) 120

$(10 \times 1 = 10)$

2. (a) Answer in **one sentence** only

(i) What is the proof technique where you assume the negation of the statement you want to prove and derive at something opposite to your assumption?

(ii) In set theory, what is the term for the set of all subsets of a given set A?

(iii) In a partially ordered set (poset), what term describes the maximum element that is less than or equal to every element in the set?

(iv) In a relation, what is it called when if a is related to b and b is related to c, then a is related to c?

(v) What is the term for a function where every element in the codomain is mapped by at least one element in the domain?

(b) Clearly write **True** or **False** for the following statements. No justification is needed for your answer.

(i) If $x \in [2,3]$ and $y \in (2,3)$ are $2 \le x \le 3$ and $2 < y < 3$ respectively, then $[1,2) \cup [2,3] = [2,3]$.

(ii) $\neg(p \rightarrow q) = p \wedge \neg q$

(iii) If $x, y \in \{1,3,5\}, R = \{(x,y) \mid x \le y\}$ then R is reflexive.

(iv) If $B = \{\{1,2\}, 3, \Phi\}$ then $|B| = 3$

(v) " If P is a lady, then P is a human being"
In this statement "P is a lady" is NECESSARY condition for "P is a human being" and "P is a human being" is SUFFICIENT condition for "P is a lady".

$(5+5=10)$

P.T.O

3. (a) Citing an example, compute $A \cup (A \cap B)$ where $A, B \subseteq N$

(b) Using membership table check if $A \cup (A \cap B) = A$

(c) Find $\bigcap_{i=1}^{n} A_i$ where $A_i = \{x \mid -i \le x \le i, \forall x \in Z\}$, $n, i \in N$

(d) Using Venn diagram show $\bigcap_{i=1}^{n} A_i$ for $n \le 3$

$(4 \times 2\frac{1}{2} = 10)$

4. Prove the following using the specified proof strategy

(a) The sum of two rational numbers is a rational number (direct proof).

(b) If $\overline{B} \subseteq \overline{A}$ then $A \subseteq B$ (method of contradiction).

(c) $n^3 + 5$ is odd, then $n \in Z$ is even (method of contraposition).

(d) $\min(a, \min(b,c)) = \min(\min(a,b), c)$ (proof by cases).

$(4 \times 2\frac{1}{2} = 10)$

5. (a) Consider the propositions, *p:* It is raining and *q:* The ground is wet. Represent the following compound propositions using logical connectives:

- The ground is wet if and only if it is raining.
- If it is not raining, then the ground is not wet.

(b) Define and provide an example of a tautology and a contradiction.

(c) Given the compound proposition $(p \wedge \neg r) \rightarrow (q \vee s)$, determine the inverse, converse, and contrapositive. Additionally, provide a truth table for the original proposition.

(d) Consider the following propositions:

- The sky is blue
- It is daytime

Explain whether the compound propositions that "During daytime the sky is blue" is satisfiable.

$(4 \times 2\frac{1}{2} = 10)$